

## BEOGRAD, Serbia

WMO: 132740

Lat: 44.7984N

Lon: 20.4648E

Elev: 132

StdP: 99.75

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -8.6       | -6.4 | -12.2                         | 1.3 | -8.0 | -10.0 | 1.6 | -4.7 | 9.2                   | 0.0  | 8.0 | 1.3  | 2.6                   | 120  | 0.404 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.8                    | 34.6            | 21.4 | 32.9 | 21.2 | 31.2 | 20.6 | 22.9                | 31.3 | 22.0 | 30.4 | 21.3 | 29.3 | 2.6                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.3                            | 15.2 | 26.8 | 19.3 | 14.3 | 25.5 | 18.6 | 13.6 | 24.6 | 68.2          | 31.5 | 65.0 | 30.2 | 62.3 | 29.4 | 26.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.3               | 6.2  | 5.2 | -10.7                      | 37.2 | 3.0                | 1.8 | -12.8                                              | 38.5 | -14.5      | 39.5 | -16.2      | 40.6 | -18.4      | 41.9 |
|                   |      |     | -11.3                      | 24.4 | 2.8                | 0.9 | -13.3                                              | 25.1 | -14.9      | 25.6 | -16.5      | 26.2 | -18.5      | 26.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 13.4        | 2.0   | 4.0   | 8.5   | 13.8  | 18.3  | 21.8  | 23.8  | 23.9  | 18.9  | 13.7  | 8.5   | 3.2   |  |  |
|                                                                        | DBStd       | 8.90        | 5.01  | 5.38  | 4.85  | 4.28  | 3.98  | 4.00  | 3.58  | 3.66  | 4.08  | 4.51  | 4.91  | 4.74  |  |  |
|                                                                        | HDD10.0     | 848         | 251   | 176   | 87    | 16    | 1     | 0     | 0     | 0     | 0     | 17    | 85    | 216   |  |  |
|                                                                        | HDD18.3     | 2386        | 506   | 401   | 306   | 144   | 52    | 13    | 3     | 43    | 153   | 294   | 468   |       |  |  |
|                                                                        | CDD10.0     | 2098        | 3     | 9     | 39    | 130   | 258   | 355   | 427   | 432   | 266   | 131   | 41    | 5     |  |  |
|                                                                        | CDD18.3     | 596         | 0     | 0     | 0     | 8     | 51    | 118   | 172   | 177   | 59    | 9     | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 5441        | 0     | 0     | 1     | 59    | 382   | 1106  | 1702  | 1718  | 427   | 44    | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 2125        | 0     | 0     | 0     | 10    | 86    | 416   | 724   | 757   | 128   | 4     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.3         | 2.5   | 2.6   | 2.6   | 2.5   | 2.2   | 2.0   | 2.0   | 1.9   | 2.1   | 2.3   | 2.7   | 2.4   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 680         | 49    | 43    | 50    | 55    | 70    | 90    | 66    | 52    | 52    | 47    | 51    | 57    |  |  |
|                                                                        | PrecMax     | 1095        | 112   | 128   | 145   | 158   | 280   | 212   | 262   | 168   | 135   | 185   | 132   | 179   |  |  |
|                                                                        | PrecMin     | 465         | 4     | 0     | 2     | 4     | 9     | 15    | 3     | 4     | 3     | 0     | 5     | 0     |  |  |
|                                                                        | PrecStd     | 139         | 29    | 30    | 33    | 26    | 46    | 49    | 53    | 37    | 33    | 37    | 27    | 37    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 15.5  | 18.6  | 22.7  | 28.0  | 31.2  | 34.5  | 37.0  | 37.5  | 33.3  | 26.7  | 22.0  | 16.6  |  |  |
|                                                                        |             | MCWB        | 10.1  | 10.5  | 12.7  | 16.4  | 19.4  | 21.6  | 21.7  | 22.0  | 20.3  | 17.9  | 14.5  | 11.5  |  |  |
|                                                                        | 2%          | DB          | 12.8  | 16.0  | 20.3  | 24.8  | 28.9  | 32.7  | 34.5  | 34.9  | 30.0  | 24.5  | 19.5  | 14.1  |  |  |
|                                                                        |             | MCWB        | 8.5   | 9.7   | 11.7  | 15.1  | 18.4  | 21.4  | 21.6  | 21.3  | 19.2  | 16.7  | 13.4  | 9.8   |  |  |
|                                                                        | 5%          | DB          | 11.0  | 13.8  | 18.3  | 22.7  | 27.3  | 31.0  | 32.6  | 32.9  | 27.6  | 22.5  | 17.5  | 11.9  |  |  |
|                                                                        |             | MCWB        | 7.7   | 8.7   | 10.9  | 13.9  | 17.9  | 20.8  | 21.3  | 21.0  | 18.5  | 15.9  | 12.5  | 8.5   |  |  |
|                                                                        | 10%         | DB          | 9.1   | 11.7  | 15.9  | 20.7  | 25.5  | 29.2  | 30.7  | 30.9  | 25.6  | 20.5  | 15.6  | 9.8   |  |  |
|                                                                        |             | MCWB        | 6.3   | 7.5   | 9.8   | 12.9  | 17.2  | 20.1  | 20.6  | 20.5  | 17.7  | 15.1  | 11.3  | 7.2   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.4  | 11.4  | 13.7  | 17.4  | 20.5  | 24.0  | 23.8  | 24.3  | 21.2  | 18.8  | 15.3  | 12.0  |  |  |
|                                                                        |             | MCDB        | 14.8  | 17.0  | 21.2  | 26.4  | 28.6  | 31.6  | 32.2  | 32.7  | 30.8  | 25.3  | 20.4  | 15.5  |  |  |
|                                                                        | 2%          | WB          | 8.9   | 10.1  | 12.3  | 15.7  | 19.3  | 22.5  | 22.7  | 22.7  | 19.9  | 17.4  | 13.9  | 10.1  |  |  |
|                                                                        |             | MCDB        | 12.1  | 15.2  | 18.8  | 23.1  | 26.9  | 30.3  | 31.2  | 31.4  | 27.8  | 22.9  | 18.6  | 13.6  |  |  |
|                                                                        | 5%          | WB          | 7.7   | 9.0   | 11.2  | 14.6  | 18.5  | 21.5  | 22.0  | 21.9  | 19.1  | 16.4  | 12.8  | 8.7   |  |  |
|                                                                        |             | MCDB        | 10.7  | 13.2  | 17.3  | 21.2  | 25.9  | 29.1  | 30.3  | 30.3  | 26.0  | 21.4  | 16.7  | 11.7  |  |  |
|                                                                        | 10%         | WB          | 6.6   | 7.9   | 10.1  | 13.6  | 17.6  | 20.6  | 21.2  | 21.1  | 18.2  | 15.4  | 11.7  | 7.2   |  |  |
|                                                                        |             | MCDB        | 9.0   | 11.4  | 15.3  | 19.4  | 24.5  | 27.9  | 29.1  | 29.1  | 24.4  | 19.8  | 15.3  | 9.5   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.9   | 6.2   | 7.8   | 8.8   | 9.4   | 9.5   | 9.8   | 10.1  | 8.9   | 8.1   | 6.3   | 4.8   |  |  |
|                                                                        |             | MCDBR       | 7.7   | 9.4   | 11.2  | 11.7  | 11.7  | 11.7  | 12.3  | 12.5  | 11.7  | 10.7  | 9.2   | 7.7   |  |  |
|                                                                        | 5% WB       | MCWBR       | 4.8   | 5.2   | 5.6   | 5.1   | 4.7   | 4.4   | 4.2   | 4.3   | 4.3   | 5.0   | 4.7   | 4.9   |  |  |
|                                                                        |             | MCWBR       | 7.2   | 8.7   | 10.3  | 10.5  | 10.9  | 10.8  | 11.0  | 11.3  | 10.7  | 9.9   | 8.4   | 7.3   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.345 | 0.371 | 0.402 | 0.458 | 0.446 | 0.441 | 0.444 | 0.454 | 0.432 | 0.396 | 0.359 | 0.342 |  |  |
|                                                                        |             | taud        | 2.336 | 2.249 | 2.216 | 2.069 | 2.167 | 2.217 | 2.205 | 2.177 | 2.206 | 2.301 | 2.361 | 2.355 |  |  |
|                                                                        | Ebn at Noon |             | 751   | 796   | 823   | 803   | 828   | 832   | 824   | 799   | 781   | 758   | 727   | 713   |  |  |
|                                                                        |             | Edh at Noon | 81    | 105   | 123   | 155   | 145   | 139   | 139   | 138   | 124   | 99    | 78    | 72    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.24  | 2.07  | 3.30  | 4.54  | 5.53  | 6.14  | 6.22  | 5.51  | 3.86  | 2.66  | 1.62  | 1.04  |  |  |
|                                                                        | RadStd      |             | 0.17  | 0.27  | 0.39  | 0.46  | 0.53  | 0.57  | 0.47  | 0.52  | 0.42  | 0.29  | 0.14  | 0.15  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.83 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (52 neighbors) |  | +0.68 | N/A     | N/A    | N/A     | +0.37 | +0.44 | -110        | -168    | +113    | +60     |

Nomenclature: See separate page